

Chief series: maximal normal series

Chief factor: part of chief series

N/L are s.t. $\nexists$ normal subgroups inside.

Regional (lg): pertaining to sufficiently large compactly generated open subgroups

Local (lg): pertaining to sufficiently small neigh. of 1

1- Normal compression

(Normal compression)

Def: H is a normal compression of a top group G if there is an injective, cont homom $\psi: G \hookrightarrow H$ with dense normal image.

Obs: When G and H are Polish groups, there is a Polish group $\tilde{H}$ s.t. ψ can be decomposed as $\beta \circ \alpha$ with $\alpha: G \hookrightarrow \tilde{H}$ a closed embedding with normal image and $\beta: \tilde{H} \twoheadrightarrow H$ a quotient map.

Thm:

Suppose G and H are t.d.l.c.s.c. groups with $\psi: G \rightarrow H$ a normal compression map. Then $\exists \tilde{H}$ t.d.l.c.s.c. group, a closed emb. $\alpha: G \hookrightarrow \tilde{H}$ with normal image and a quot map $\beta: \tilde{H} \twoheadrightarrow H$ s.t.

(i) $\psi = \beta \circ \alpha$;

(ii) $\tilde{H}/\alpha(G)$ is compact;

(iv) $\tilde{H} = \alpha(G) \text{Ker}(\beta)$.

(iii) $\text{Ker}(\beta)$ is discrete, lies in the FC-center of $\tilde{H}$ and centralizes $\alpha(G)$;

Thm:

Suppose G and H are l.c.s.c. groups with H a normal compression of G .

(i) G is amenable iff H is amenable.

(ii) If G and H are t.d., then G is elementary with decomposition rank ∞ iff H is elementary with decomposition rank ∞ .

def: (Quasidiscrete) G

A Polish group is quasi-discrete if the quasi-center

$$QZ(G) := \{g \in G \mid C_G(g) \text{ is open}\}$$

is open.

Obs: The property of being discrete is not preserved under normal compressions, but quasi-discreteness is preserved up to metabelian factors.

Thm:

Suppose that G and H are l.c.s.c. groups with H a normal compression of G .

(i) if H is quasi-discrete $\Rightarrow G/QZ(G)$ is metabelian.

(ii) if G is quasi-discrete $\Rightarrow H/QZ(H)$ is metabelian.

2/Generalized direct products

Def: (Quasi-direct factorization)

For G a topological group and $\mathcal{J}$ a collection of closed normal subgroups, we define set $G_{\mathcal{J}} = \overline{\langle \bigcup_{N \in \mathcal{J}} N \rangle}$. A collection $\mathcal{A}$ of closed normal subgroups is a quasi-direct factorization of G if $G_{\mathcal{A}} = G$, and $\mathcal{A}$ has the following independence property:

$$\forall X \subseteq \mathcal{P}(\mathcal{A}): \bigcap X = \emptyset \Rightarrow \bigcap_{A \in X} G_A = \{1\}.$$

In such a case, $(G, \mathcal{A})$ is said to be a quasi-product. We say that G is a quasi-product when the factorisation is implicit

Def (Local direct product)

Suppose that $(G_i)_{i \in I}$ is a set of top. groups and that there is a distinguished open subgrp. $Q_i \leq G_i$ for each $i \in I$. The local product of $(G_i)_{i \in I}$ over $(Q_i)_{i \in I}$ is denoted by $\bigoplus_{i \in I} (G_i, Q_i)$ and defined to be

$$\{f: I \rightarrow \prod_{i \in I} G_i \mid f(i) \in G_i, \text{ and } f(i) \in Q_i \text{ for all but finitely many } i\}$$

with component-wise multiplication and the group top. s.t. $\prod_{i \in I} Q_i$

with the prod top cont. embeds as an open subgrp of $\bigoplus_{i \in I} (G_i, Q_i)$

Def (Local direct model)

Let G be a l.c. group with Δ a quasi-direct factorisation of G .

A local direct model for (G, Δ) is a cont homom. $\phi: \bigoplus_{\Delta} (G) \rightarrow G$,

where $\bigoplus_{\Delta} (G) := \bigoplus_{N \in \Delta} (N, N \cap \mathcal{Q})$ for some open subgroup $\mathcal{Q}$ of G

s.t. $\bigoplus_{\Delta} (G)$ is l.c. and $\phi|_N = \text{id}_N$ for all $N \in \Delta$.

Obs: Local direct models for quasi-products, provided they exist, enjoy a universal property and are in particular unique.

Thrm:

If (G, Δ) is a t.d.l.c. quasi-product, then (G, Δ) admits a local direct model.

Def: (Strictly semisimple type)

A Polish group G is of strictly semisimple type if it is quasi-product of non-abelian topologically simple groups.

Thm:

For G a l.c.s.c group of strict semisimple type

$$G \simeq D \times \prod_{i \in I} C_i \times \prod_{j=1}^m S_j$$

with D a t.d. l.c.s.c. group of strict semisimple type, each C_i is a topologically simple compact connected Lie group, and each S_j a non-compact ~~topologically~~ topologically simple connected Lie group.

Coro:

Suppose G is an l.c.s.c. group of strict semisimple type and let Λ be the set of non-abelian top. simple closed normal subgroups of G . Then (G, Λ) is a quasi-product that admits a local direct model.

3/Chief factors and blocks.

Def: (Normal factor/Chief factor)

A closed normal factor of a group G is a pair K/L of closed normal subgroups of G s.t. $L \leq K$. It is a chief factor if in addition, $L < K$ and there is no closed $M \triangleleft G$ s.t. $L < M < K$.

Def: (Associated normal factors)

Given a top. group G , we say that the closed normal factors K_1/L_1 and K_2/L_2 are associated if the following eq holds:

$$\overline{K_1 L_2} = \overline{K_2 L_1}$$

$$K_1 \cap \overline{L_1 L_2} = L_1$$

$$K_2 \cap \overline{L_1 L_2} = L_2$$

Obs: This is an equivalence relation when restricted to the ~~non-abel~~ ~~non-abel~~ non-abelian chief factors.

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Def: (Chief Block)

A chief block is an association class of non-abelian chief factors